Question-4.10.24

EE25BTECH11022 - sankeerthan

Find the equations of the lines through the point of intersection of the line $x - y + 1 = 0$ and $2x - 3y + 5 = 0$ and whose distance from the point $(3, 2)$ is $\frac{7}{5}$.

Solution

Given lines are

$$\begin{pmatrix}1 & -1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}+1=0, \begin{pmatrix}2 & -3\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}+5=0 \tag{1}$$

$$\begin{pmatrix}1 & -1\\2 & -3\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}-1\\-5\end{pmatrix} \tag{2}$$

Inverse of coefficient matrix

$$\det = 1\times(-3)-(-1)\times 2=-3+2=-1 \tag{3}$$

$$A^{-1}=\frac{1}{-1}\begin{pmatrix}-3 & 1\\-2 & 1\end{pmatrix}=\begin{pmatrix}3 & -1\\2 & -1\end{pmatrix} \tag{4}$$

Intersection point of given two lines is

$$\begin{pmatrix}x_0\\y_0\end{pmatrix}=A^{-1}\begin{pmatrix}-1\\-5\end{pmatrix}=\begin{pmatrix}3 & -1\\2 & -1\end{pmatrix}\begin{pmatrix}-1\\-5\end{pmatrix}=\begin{pmatrix}2\\3\end{pmatrix} \tag{5}$$

Solution

General line through (x_0, y_0) with normal vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is

$$\begin{pmatrix} a & b \end{pmatrix} \left(\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) = 0 \tag{6}$$

Distance from point $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ to line is

$$Distance = \frac{\left| \begin{pmatrix} a & b \end{pmatrix} \left(\begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) \right|}{\sqrt{\begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}}} = \frac{7}{5} \tag{7}$$

$$\frac{\left| \begin{pmatrix} a & b \end{pmatrix} \left(\begin{pmatrix} 3-2 \\ 2-3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) \right|}{\sqrt{\begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}}} = \frac{7}{5} \tag{8}$$

Solution

Let t $= \frac{a}{b} \quad (b \neq 0)$

$$\frac{|t-1|}{\sqrt{t^2+1}} = \frac{7}{5} \tag{10}$$

$$\tag{11}$$

Squaring on both sides

$$25(t-1)^2 = 49(t^2+1) \tag{12}$$

$$25(t^2-2t+1) = 49t^2+49 \tag{13}$$

$$25t^2-50t+25 = 49t^2+49 \tag{14}$$

$$24t^2+50t+24 = 0 \tag{15}$$

$$12t^2+25t+12 = 0 \tag{16}$$

$$t = \frac{-25 \pm \sqrt{25^2 - 4 \times 12 \times 12}}{2 \times 12} = \frac{-25 \pm 7}{24} \tag{17}$$

Solutions $\quad t_1 = -\frac{3}{4}, t_2 = -1$

Solution

Normal vectors $(let b = 1)$:

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \frac{-3}{4} \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \tag{18}$$

Equation constants c = -a$\times$2 $- b \times 3$

$$c_1 = (-2) \times \left(-\frac{3}{4}\right) - 3 \times 1 = \frac{3}{2} - 3 = \frac{-3}{2} \tag{19}$$

$$c_2 = (-2) \times (-1) - 3 \times 1 = 2 - 3 = -1 \tag{20}$$

required equations of line are

$$\begin{pmatrix} \frac{-3}{4} & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - \frac{3}{2} = 0 \tag{21}$$

$$\begin{pmatrix} -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - 1 = 0 \tag{22}$$

$$\tag{23}$$

graph

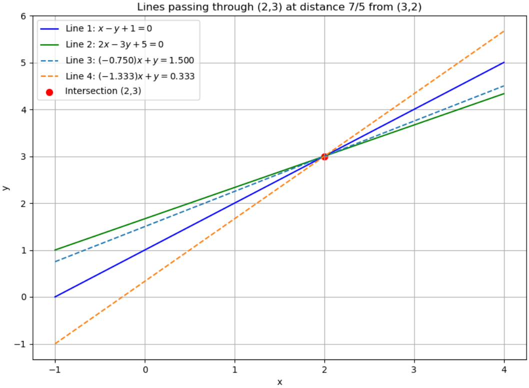

Figure:

C-Code

```c
// code.c
#include <stdio.h>
#include <math.h>

// Solve quadratic ax^2 + bx + c = 0
void solve_quadratic(double a, double b, double c, double *r1,
    double *r2) {
    double d = b*b - 4*a*c;
    if (d < 0) {
        *r1 = NAN;
        *r2 = NAN;
        return;
    }
    *r1 = (-b + sqrt(d)) / (2*a);
    *r2 = (-b - sqrt(d)) / (2*a);
}
```

C-Code

```c
void get_line_params(double *a1, double *a2, double *c1, double *
    c2) {
    double A = 1.0, B = 25.0/12.0, C = 1.0;
    double r1, r2;
    solve_quadratic(A, B, C, &r1, &r2);

    a1[0] = r1;
    a2[0] = r2;

    // Constants c = 2a + 3, intersection point fixed at (2,3)
    c1[0] = 2*r1 + 3;
    c2[0] = 2*r2 + 3;
}
```

Python-Code

```python
# Code by GVV Sharma
# Modified for Problem Solution
# Released under GNU GPL
# Calculating area enclosed between curves
# call.py
import ctypes
import numpy as np
import os

# Load shared library
lib = ctypes.CDLL(os.path.abspath('./code.so'))
```

Python-Code

```
a1 = ctypes.c_double()
a2 = ctypes.c_double()
c1 = ctypes.c_double()
c2 = ctypes.c_double()

# call C function
lib.get_line_params(ctypes.byref(a1),ctypes.byref(a2),ctypes.
    byref(c1),ctypes.byref(c2))

# Extract values
roots = [a1.value, a2.value]
cs = [c1.value, c2.value]

# Save to numpy for plot
np.save('roots.npy', np.array(roots))
np.save('cs.npy', np.array(cs))
print("Roots (a1, a2):", roots)
print("Constants (c1, c2):", cs)
```

Python-Code

```
import numpy as np
import matplotlib.pyplot as plt

roots = np.load('roots.npy')
cs = np.load('cs.npy')

x = np.linspace(-1, 4, 400)

plt.figure(figsize=(10, 7))

# Original lines
# Line 1: x - y + 1 = 0 => y = x + 1
y1 = x + 1
plt.plot(x, y1, label='Line 1: $x - y + 1=0$', color='blue')

# Line 2: 2x - 3y + 5 = 0 => y = (2x + 5)/3
y2 = (2*x + 5)/3
plt.plot(x, y2, label='Line 2: $2x - 3y + 5=0$', color='green')
```

Python-Code

```python
# Required lines (Line 3 and Line 4)
for i in range(2):
    a = roots[i]
    c = cs[i]
    # Equation: a x + y = c => y = -a x + c
    y = -a * x + c
    plt.plot(x, y, label=f'Line {i+3}: $({a:.3f})x + y = {c:.3f}$', linestyle='--')

# Plot intersection point (2,3)
plt.scatter(2, 3, color='red', s=50, label='Intersection (2,3)')

plt.xlabel('x')
plt.ylabel('y')
plt.title('Lines passing through (2,3)at distance 7/5 from(3,2)')
plt.legend()
plt.grid(True)
plt.axis([-1, 4, -2, 6])
plt.show()
```